LEF1 Ectodermal Dysplasia with Limb Malformations — Comprehensive Disease Report

Disease: LEF1 Ectodermal Dysplasia with Limb Malformations **Preferred clinical name:** Ectodermal dysplasia 17, with or without limb malformations (ECTD17) **MONDO:** MONDO:0979228 · **OMIM:** 621224 · **Causal gene:** *LEF1* (OMIM 153245; HGNC:6551) **Category:** Mendelian, ultra-rare

---

Summary

LEF1 Ectodermal Dysplasia with Limb Malformations (ECTD17; MONDO:0979228; OMIM 621224) is an ultra-rare Mendelian multisystem developmental disorder caused by pathogenic germline variants in *LEF1*, the HMG-box transcription factor that serves as the nuclear effector of canonical WNT/β-catenin signaling. The disorder was first delineated as a distinct clinical entity by Dufour and colleagues in 2022, who described 12 affected individuals from 5 families and established *LEF1* as the causal gene through segregation and functional assays (PMID: 35583550). The cardinal features combine **ectodermal dysplasia** (sparse eyebrows/hair, oligodontia and other dental anomalies, dry skin, nail dystrophy, hypoplastic nipples/breast tissue) with **limb malformations** (radial ray defects, polydactyly, and split-hand/split-foot malformation).

The central mechanistic insight of this investigation is that **both inheritance pattern and severity are determined by the variant class and its position within the LEF1 protein**. Monoallelic variants that cause haploinsufficiency or loss of DNA binding produce autosomal dominant, mild-to-moderate disease, whereas biallelic variants that abolish β-catenin binding produce autosomal recessive, severe disease. This genotype–severity relationship is grounded in the protein's domain architecture: the N-terminal β-catenin–binding domain (residues 1–62) and the C-terminal HMG-box DNA-binding domain (residues 299–367) define two separable loss-of-function mechanisms. LEF1 is among the most loss-of-function–intolerant genes in the genome (gnomAD pLI = 0.998, LOEUF = 0.212), independently corroborating a haploinsufficiency mechanism for the dominant form.

Mechanistically, LEF1 is required for WNT-dependent epithelial–mesenchymal signaling that drives the development of ectodermal appendages (teeth, hair, whiskers, mammary glands) and the patterning of the limb. The *Lef1*-null mouse recapitulates the ectodermal and mammary components of the human phenotype — absent teeth, hair, vibrissae and mammary glands — but notably **not** the limb-reduction phenotype, an allele/dosage-specific observation that highlights the difference between complete gene ablation (mouse) and the specific partial/domain-selective losses seen in human patients. There is no disease-modifying therapy; management is supportive and multidisciplinary.

Section 1 — Disease Information

Overview. ECTD17 is a congenital, developmental syndrome in which disrupted canonical WNT signaling impairs the formation of ectodermal appendages and limb structures. It belongs to the broad family of ectodermal dysplasias (ED), which are defined by abnormalities of two or more ectodermal derivatives (hair, teeth, nails, sweat glands), but is distinguished by the co-occurrence of radial-ray / split-hand-foot limb malformations.

Key identifiers (verified via EBI OLS / MONDO cross-reference query for MONDO:0979228):

Resource	Identifier
MONDO	MONDO:0979228 — "ectodermal dysplasia 17 with or without limb malformations"
OMIM	621224 (phenotype); 153245 (*LEF1* gene)
MedGen	C1876515 (CUI 1876515)
UMLS	C6012731
GARD	0028106
Gene	*LEF1* (HGNC:6551)

Synonyms / alternative names: Ectodermal dysplasia 17 with or without limb malformations; ECTD17; LEF1-related ectodermal dysplasia with limb malformations; LEF1 ectodermal dysplasia–limb malformation syndrome.

Information source. Disease-level evidence is derived predominantly from a single aggregated case series (Dufour et al., 2022, 12 individuals) plus curated ontology/database resources (OMIM, MONDO, HPO, ClinVar, gnomAD, UniProt, MGI). This is **not** EHR/population-derived; it is an aggregation of individual-patient reports consolidated into database annotations. (Finding F004.)

Section 2 — Etiology

Primary cause. ECTD17 is a monogenic disorder caused by pathogenic germline variants in *LEF1*. There is no known environmental or infectious contribution. (Finding F001.)

Genetic risk / causal factors. - **Monoallelic (heterozygous) variants** → autosomal dominant, mild-to-moderate disease, via **haploinsufficiency** or **loss of DNA binding**. - **Biallelic variants** (observed in a consanguineous family) → autosomal recessive, severe disease, via **loss of β-catenin binding**.

Dufour et al. summarized the two-mechanism model directly: *"Our functional data indicate that 2 molecular mechanisms are at play: haploinsufficiency or loss of DNA binding are responsible for a mild to moderate phenotype, whereas loss of β-catenin binding caused by biallelic variants is associated with a severe phenotype."* (PMID: 35583550) (Finding F001.)

Environmental risk factors / protective factors / gene–environment interactions. None established. Given the monogenic, developmental nature of the disorder, there is no literature identifying environmental triggers, protective alleles, lifestyle factors, or GxE interactions. **Consanguinity** is a relevant demographic factor for the recessive, severe form (the biallelic case arose in a consanguineous family). (Finding F001.)

Section 3 — Phenotypes

Phenotype frequencies below are from official HPO annotations for OMIM:621224, sourced to Dufour et al. 2022 (PMID: 35583550). The authors summarized: *"The phenotypic spectrum includes various limb malformations, such as radial ray defects, polydactyly or split hand/foot, and ectodermal dysplasia."* (Finding F005.)

Limb malformations (physical manifestations)

Phenotype	Frequency	Suggested HPO term
Split hand	3/12	HP:0001171 (Split hand)
Short thumb	4/11	HP:0009778 (Short thumb)
Small thenar eminence	4/12	HP:0010486 (Thenar muscle atrophy/hypoplasia)
Radial ray deficiency	1/12	HP:0002984 (Hypoplasia of the radius / radial ray)
Split foot	1/12	HP:0001839 (Split foot)
Preaxial polydactyly (hand)	case(s)	HP:0001177 (Preaxial hand polydactyly)
Toe syndactyly	case(s)	HP:0001770 (Toe syndactyly)

Ectodermal features

Phenotype	Frequency	Suggested HPO term
Oligodontia	5/14	HP:0000677 (Oligodontia)
Sparse eyebrows	6/12	HP:0045025 (Sparse eyebrow)
Dry skin	5/13	HP:0000958 (Dry skin)
Eczematoid dermatitis	3/12	HP:0000964 (Eczema)
Abnormal nails	3/12	HP:0001597 (Abnormality of the nail)
Taurodontia / microdontia / delayed eruption	case(s)	HP:0000679 / HP:0000691 / HP:0000684
Hypohidrosis	1/12	HP:0000966 (Hypohidrosis)

Mammary features

Phenotype	Frequency	Suggested HPO term
Hypoplastic nipples	4/10	HP:0002557 (Hypoplastic nipples)
Breast hypoplasia	2/10	HP:0003187 (Breast hypoplasia)

The mammary phenotype is mechanistically notable because it directly mirrors the *Lef1*-null mouse (absent mammary gland), consistent with LEF1's known role in mammary bud formation (see Sections 6 and 15).

Other / variable features

Phenotype	Frequency	Suggested HPO term
Mild intellectual disability	1/12	HP:0001256
Autism spectrum disorder	1/12	HP:0000729
Patent ductus arteriosus	1/12	HP:0001643
Cryptorchidism	case(s)	HP:0000028
Short stature	1/3	HP:0004322

Phenotype characteristics. *Onset:* congenital/neonatal (limb and ectodermal structures are developmentally determined). *Severity:* variable, and strongly genotype-dependent (mild-to-moderate in monoallelic AD cases; severe in the biallelic AR case). *Progression:* structural malformations are stable (non-progressive); ectodermal features (dry skin, dental anomalies) are lifelong. *Quality-of-life impact:* driven mainly by hand/foot function (dexterity, grip), dental function/aesthetics, and skin barrier issues; disease-specific QoL instruments have not been applied in the literature. (Finding F005.)

Section 4 — Genetic / Molecular Information

Causal gene. *LEF1* (lymphoid enhancer-binding factor 1), OMIM 153245, HGNC:6551, located at chromosome **4q25** (chr4:~108.97–109.09 Mb, GRCh37). Canonical transcript NM_016269.5; protein UniProt Q9UJU2 (399 aa). (Findings F006, F007.)

Constraint / dosage sensitivity. gnomAD constraint metrics (GRCh37, ENSG00000138795) place LEF1 among the most loss-of-function–intolerant genes:

Metric	Value	Interpretation
pLI	0.998	Extreme LoF intolerance
LOEUF (oe_lof_upper)	0.212	Strong constraint
Observed/expected LoF	1 / 22.4 (o/e = 0.045)	Near-complete depletion of LoF alleles
LoF Z-score	4.19	Highly significant
Missense Z-score	1.57	Mild missense constraint

This independent population-genetic signal supports haploinsufficiency as the dominant-disease mechanism — consistent with the functional demonstration that *"haploinsufficiency or loss of DNA binding are responsible for a mild to moderate phenotype"* (PMID: 35583550). (Finding F006.)

Pathogenic variants (ClinVar, transcript NM_016269.5).

Variant (cDNA)	Protein	Type	Classification	Association
c.856del	p.His286fs	Frameshift	**Pathogenic**	ECTD17
c.469dup	p.Leu157fs	Frameshift	Conflicting	ECTD17
c.811G>T	p.Glu271Ter	Nonsense	VUS	—
Large 4q CNVs encompassing *LEF1*	—	Structural (deletion/CNV)	Pathogenic / Likely pathogenic	—

Variant classes reported for the disorder include **frameshift, nonsense, and missense** alleles plus large structural CNVs. Both of the truncating frameshifts (p.Leu157fs at residue 157, p.His286fs at residue 286) terminate the protein **before** the C-terminal HMG box (residues 299–367), removing the DNA-binding domain — the structural correlate of the "loss of DNA binding" mechanism. (Findings F006, F007.)

Domain architecture and genotype–severity mapping (UniProt Q9UJU2).

```
LEF1 protein (399 aa)
 │
 ├─ aa 1–62    N-terminal β-CATENIN (CTNNB1)-BINDING DOMAIN
 │              → biallelic loss here = loss of β-catenin binding = SEVERE (AR)
 │
 ├─ central    Intrinsically disordered, context-dependent regulatory region
 │
 └─ aa 299–367 C-terminal HMG-BOX DNA-BINDING DOMAIN (DNA bending)
                → truncation/loss here = loss of DNA binding = MILD–MODERATE (AD)
```

Dufour et al.: *"whereas loss of β-catenin binding caused by biallelic variants is associated with a severe phenotype."* (PMID: 35583550) The N-terminal CTNNB1-binding domain (aa 1–62) is the structural correlate of this severe recessive mechanism. (Finding F007.)

Relevant GO annotations for LEF1 (Q9UJU2): - Molecular function: beta-catenin binding (GO:0008013); gamma-catenin binding (GO:0045295); DNA binding, bending (GO:0008301). - Biological process: canonical Wnt signaling (GO:0060070); positive regulation of odontoblast differentiation (GO:1901331); secondary palate development (GO:0062009); osteoblast differentiation (GO:0001649); epithelial–mesenchymal transition (GO:0001837 / GO:0010718).

Modifier genes / epigenetics. Not established for this disorder. No disease-specific methylation or histone-modification signatures have been reported.

Chromosomal abnormalities. Large 4q deletions/CNVs encompassing *LEF1* are classified Pathogenic/Likely pathogenic and represent a structural route to LEF1 haploinsufficiency. (Finding F006.)

Section 5 — Environmental Information

No environmental, lifestyle, or infectious factors are implicated. ECTD17 is a monogenic developmental disorder; environmental factors (toxins, radiation, occupational exposures, smoking/diet/alcohol) and infectious agents are **not applicable**. The only non-genetic epidemiological modifier of relevance is **consanguinity**, which raises the risk of the biallelic (recessive, severe) form. (Finding F001.)

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. A germline *LEF1* variant (frameshift/nonsense/missense/CNV, mono- or biallelic) **leads to** reduced LEF1 dosage and/or a LEF1 protein lacking a functional domain.
2. **Branch A (monoallelic):** haploinsufficiency or a variant that removes/impairs the HMG-box DNA-binding domain (aa 299–367) **results in** a quantitative reduction in functional LEF1 available to act on WNT target enhancers → *mild-to-moderate* disease.
3. **Branch B (biallelic):** variants abolishing the N-terminal β-catenin–binding domain (aa 1–62) **result in** a LEF1 that cannot be co-activated by β-catenin → near-complete loss of WNT transcriptional output at LEF1 sites → *severe* disease.
4. Reduced LEF1/β-catenin transcriptional complex activity **leads to** altered expression of canonical WNT target genes (demonstrated by RNA-seq showing altered WNT signaling in patient-derived material). (Finding F001.)
5. Impaired WNT transcriptional output **disrupts** WNT-dependent **epithelial–mesenchymal signaling** required for the induction and morphogenesis of ectodermal appendages (hair follicles, teeth, nails, sweat glands, mammary buds) — *inferred* from LEF1's established role as a WNT readout in these tissues. (Findings F002, F008.)
6. In parallel, impaired WNT-dependent limb-bud patterning (radial ray / AER–ZPA programs) **results in** radial ray defects, polydactyly, and split-hand/split-foot malformation — the step that is *allele/dosage-specific in humans and not reproduced by complete gene knockout in mouse*. (Finding F008.)
7. The combined ectodermal and limb developmental failures **manifest clinically** as ECTD17.

Supporting detail by category

Molecular pathways. Canonical **WNT/β-catenin** signaling. LEF1 is an HMG-box TCF/LEF-family transcription factor that binds β-catenin and acts as the nuclear transcriptional effector of the pathway. Across multiple independent systems LEF1 is a direct WNT readout: in hair-follicle dermal papilla cells, WNT+FGF activation increased *LEF1* 3.8-fold ($p < 0.01$) (PMID: 42283394); bacterial extracellular vesicles induced the canonical Wnt readouts LEF1 and AXIN2 in dermal papilla cells and promoted hair-shaft elongation (PMID: 42316873). (Finding F002.)

Cellular processes. WNT-driven epithelial–mesenchymal interaction/induction; cell-fate specification and morphogenesis of ectodermal placodes and limb mesenchyme; epithelial–mesenchymal transition (GO:0001837). LEF1 is expressed in the signaling mesenchymal compartments — *"the dermal papilla of hair and whisker follicles, the dental papilla of teeth, and a subset of mesenchymal cells in pubescent mammary glands"* (PMID: 9786416). (Finding F008.)

Protein dysfunction. Loss of function via (i) truncation removing the HMG box (loss of DNA binding) or (ii) loss of the N-terminal β-catenin–binding domain (loss of co-activator recruitment). These are **loss-of-function** / **hypomorphic** mechanisms, not gain-of-function. Haploinsufficiency is supported by extreme gnomAD LoF intolerance (pLI 0.998). (Findings F006, F007.)

Immune / metabolic / tissue-damage mechanisms. Not primary to this developmental disorder. (LEF1/TCF biology is prominent in T-cell memory and oncogenesis in other contexts, but these are not the pathomechanism of ECTD17.)

Molecular profiling. Dufour et al. performed RNA-seq (plus immunofluorescence, immunoblotting, and pulldown) demonstrating altered WNT signaling as the downstream consequence of the variants. (Finding F001.)

Context — ectodermal dysplasia as a WNT/EDA pathway disease

The placement of ECTD17 within WNT-pathway ED is reinforced by the broader literature: *"Molecular pathogenesis of ED involves mutations of genes encoding key proteins of major developmental pathways, including ectodysplasin (EDA) and wingless-type (WNT) pathways."* (PMID: 34573371) (Findings F002, F003.)

Suggested ontology terms

- **GO biological process:** GO:0060070 (canonical Wnt signaling), GO:1901331 (positive regulation of odontoblast differentiation), GO:0062009 (secondary palate development), GO:0001837 (epithelial–mesenchymal transition), GO:0042476 (odontogenesis), GO:0001942 (hair follicle development), GO:0030879 (mammary gland development).
- **GO molecular function:** GO:0008013 (beta-catenin binding), GO:0008301 (DNA binding, bending).
- **CL cell types:** CL:0002332 (dermal papilla cell), CL:0000062 (odontoblast)/dental papilla cell, mammary mesenchymal cells, limb-bud mesenchymal cells.

Section 7 — Anatomical Structures Affected

Organ / system level. - **Integumentary system (primary):** skin (dry skin, eczematoid dermatitis), hair (sparse eyebrows/hair), nails (dystrophy), sweat glands (hypohidrosis). UBERON:0002097 (skin of body), UBERON:0001037 (strand of hair), UBERON:0001705 (nail), UBERON:0001820 (sweat gland). - **Dentition:** teeth (oligodontia, taurodontia, microdontia, delayed eruption). UBERON:0001091 (tooth). - **Musculoskeletal / limbs:** hand/foot skeleton, radial ray, thenar musculature. UBERON:0002398 (manus), UBERON:0002387 (pes), UBERON:0002102 (forelimb), UBERON:0003823 (thenar / manus region). - **Mammary:** nipple and breast tissue (hypoplastic nipples, breast hypoplasia). UBERON:0001911 (mammary gland), UBERON:0002030 (nipple). - **Secondary / variable:** cardiovascular (PDA — UBERON:0002061 ductus arteriosus), CNS (mild ID/ASD), reproductive (cryptorchidism).

Tissue / cell level. Primarily **epithelial** derivatives of ectoderm and the **mesenchymal** signaling populations that instruct them (dermal papilla, dental papilla, mammary mesenchyme, limb-bud mesenchyme). Cell Ontology: CL:0002332 (dermal papilla cell), dental papilla cells, mammary mesenchymal cells.

Subcellular level. LEF1 is a transcription factor; its functional compartment is the **nucleus** (GO:0005634), acting at **chromatin** (GO:0000785) within the β-catenin/LEF1 transcriptional complex.

Localization / lateralization. Limb anomalies are typically **bilateral** but may be **asymmetric** (variable between hands and feet); ectodermal features are generalized/bilateral.

Section 8 — Temporal Development

- **Onset:** congenital. Both limb malformations and ectodermal appendage defects are determined during embryonic/fetal development; dental features become evident through infancy/childhood as dentition develops. (Finding F005.)
- **Onset pattern:** developmental (non-acute); features are present from birth or emerge as the relevant structures normally develop.
- **Progression:** structural malformations (limbs, mammary hypoplasia) are **stable/non-progressive**; ectodermal features (dry skin, dental anomalies, nail dystrophy) are **lifelong/chronic** but non-progressive. No disease staging exists.

- **Disease course / duration:** chronic, lifelong; not life-limiting in the reported cohort apart from variable complications.
- **Remission:** none (structural/developmental disorder). No spontaneous or treatment-induced remission.
- **Critical period:** the window of embryonic limb-bud patterning and ectodermal placode induction — the only period during which the primary malformations could, in principle, be modified (not currently actionable).

Section 9 — Inheritance and Population

Epidemiology. Prevalence and incidence are **unknown**; the disorder is ultra-rare, with ~12 individuals reported to date. For epidemiological context, the limb component overlaps split-hand/split-foot malformation (SHFM), which *"is a rare condition that occurs in 1 in 8500-25,000 newborns and accounts for 15% of all limb reduction defects."* (PMID: 30622331) — but LEF1-specific figures are not established. (Finding F003.)

Inheritance. Both autosomal dominant and autosomal recessive, depending on variant class: - **Autosomal dominant** — monoallelic haploinsufficiency or loss of DNA binding (11 affected individuals across 4 unrelated families in the original report). - **Autosomal recessive** — biallelic loss of β-catenin binding (1 individual, consanguineous family). (Findings F001, F005.)

Penetrance / expressivity. Expressivity is **variable** (range of limb and ectodermal severity, including incomplete feature sets per the "with or without limb malformations" nomenclature). Precise penetrance figures are not quantified given the small cohort.

Other genetic parameters. Genetic anticipation — not applicable (not a repeat-expansion disorder). Germline mosaicism — not reported. Founder effects — none known. **Consanguinity** — contributory to the recessive severe form. Carrier frequency — not established; given extreme LoF intolerance, LoF alleles are very rare in gnomAD (observed LoF = 1). (Finding F006.)

Population demographics. No ethnic predilection, geographic clustering, sex bias, or variant-specific geographic distribution has been established in the small reported cohort.

Section 10 — Diagnostics

Clinical recognition. Diagnosis is clinical-plus-molecular: recognition of the combination of ectodermal dysplasia features (hypodontia/oligodontia, sparse hair/eyebrows, dry skin, nail/sweat-gland anomalies, hypoplastic nipples) with limb malformations (radial ray defects, polydactyly, split-hand/foot), followed by confirmatory genetic testing.

Imaging. Hand/foot radiographs to characterize radial ray deficiency, split-hand/foot, polydactyly, and thenar hypoplasia; dental radiographs/panoramic imaging for oligodontia and taurodontia. Echocardiography where PDA or other cardiac involvement is suspected.

Genetic testing (recommended, definitive). - **Whole-exome sequencing (WES)** or a targeted **ectodermal dysplasia / limb-malformation gene panel** including *LEF1* is the primary approach, given phenotypic overlap with other WNT/ectodermal loci. - **Single-gene *LEF1* sequencing** confirms point/indel variants (NM_016269.5). - **Chromosomal microarray (CMA) / WGS** to detect large **4q CNVs** encompassing *LEF1* when sequencing is negative but dosage loss is suspected. (Finding F006.) - Variant interpretation per ACMG/AMP; truncating variants and whole-gene deletions are readily classified given strong LoF intolerance, while missense variants may require functional follow-up (β-catenin binding / DNA binding assays, as in the original report).

Differential diagnosis. Other WNT/ectodermal and SHFM loci must be distinguished:

Gene / locus	Overlapping features	Reference
TP63 (EEC/SHFM4)	SHFM + hypodontia	PMID: 31420900, PMID: 42110406, PMID: 33294441
WNT10B, *DLX5*, *LRP6*, *UBA2*	SHFM	PMID: 31332306
10q24 duplication; 17p13.3 (*BHLHA9*)	SHFM	PMID: 30622331, PMID: 31332306
EDA/*EDAR*/*WNT10A*	Ectodermal dysplasia (hypohidrotic)	PMID: 34573371

"SHFM and hypodontia are prominent clinical manifestations of TP63-related disorders." (PMID: 31420900) — making *TP63* the key clinical mimic. (Finding F003.)

Screening. Cascade genetic testing of at-risk relatives once a familial variant is identified; prenatal/preimplantation genetic testing is feasible for known familial variants. No population newborn screening exists.

Section 11 — Outcome / Prognosis

- **Survival / mortality:** The disorder is generally **not life-limiting**; no disease-specific mortality figures are reported, and life expectancy is expected to be normal in typical (mild-to-moderate) cases. Severe (biallelic) disease and associated anomalies (e.g., PDA) could add complication-dependent risk, but data are limited to single cases.
- **Morbidity / disability:** Driven by **hand/foot function** (grip, dexterity, fine motor tasks) from radial ray/split-hand-foot defects, **dental morbidity** (oligodontia affecting mastication, speech, aesthetics), and **skin/thermoregulation** issues (dry skin, hypohidrosis).
- **Complications:** dental complications, skin barrier/eczematoid dermatitis, and — in individual cases — cardiac (PDA), neurodevelopmental (mild ID/ASD), and genitourinary (cryptorchidism) findings.
- **Prognostic factors:** the strongest prognostic determinant is **genotype** — biallelic β-catenin-binding loss predicts severe disease, whereas monoallelic haploinsufficiency/DNA-binding loss predicts milder disease. (Findings F001, F007.)
- **Quality of life:** formal QoL instruments (EQ-5D, SF-36, PROMIS) have not been applied; impact is inferred from functional/aesthetic domains.

Section 12 — Treatment

There is **no disease-modifying or curative therapy**. Management is **supportive and multidisciplinary**, directed at individual manifestations:

Domain	Intervention	Suggested NCIT term
Dental	Restorative/prosthodontic care, dental implants, orthodontics for oligodontia	NCIT:C15300 (Dental Procedure)
Limb	Reconstructive/orthopedic hand-foot surgery for SHFM/polydactyly; occupational therapy	NCIT:C15329 (Orthopedic Surgery)
Skin	Emollients/barrier care; dermatologic management of eczematoid dermatitis	NCIT:C15733 (Supportive Care)
Thermoregulation	Heat-avoidance / environmental measures for hypohidrosis	NCIT:C15733 (Supportive Care)
Rehabilitation	Physical / occupational therapy; speech therapy if dentition affects articulation	NCIT:C15351 (Rehabilitation Therapy)
Cardiac	PDA closure if clinically indicated	NCIT:C50743 (Ductus Arteriosus Repair)
Genetic	Genetic counseling for family planning	NCIT:C15417 (Genetic Counseling)

- **Pharmacotherapy / pharmacogenomics:** none specific. No targeted WNT-modulating therapy is approved or in trials for this disorder.
- **Advanced therapeutics (gene/cell/RNA/targeted/immuno):** none in clinical use or trials for ECTD17; remain theoretical. LEF1 haploinsufficiency is mechanistically a candidate for dosage-restoration approaches, but the critical developmental window largely precedes diagnosis.
- **Experimental:** no registered ClinicalTrials.gov interventions specific to this disorder (ultra-rare, recently described).

Section 13 — Prevention

- **Primary prevention:** not possible for a congenital monogenic disorder other than through **reproductive genetic counseling**. For families with a known *LEF1* variant, options include prenatal diagnosis and preimplantation genetic testing.

- **Secondary prevention:** early multidisciplinary assessment (dental, orthopedic, dermatologic, cardiac) to anticipate and manage complications; cascade testing of relatives.
- **Tertiary prevention:** optimizing dental, orthopedic, and skin care to prevent secondary functional decline.
- **Counseling:** genetic counseling is central — conveying the variant-class-dependent inheritance (AD vs AR) and recurrence risks, and the elevated risk of the severe recessive form with **consanguinity**. (Findings F001, F007.)
- **Immunization / public health / environmental interventions:** not applicable.

Section 14 — Other Species / Natural Disease

- **Taxonomy / orthologs:** Mouse *Lef1* (NCBI Gene 16842; MGI:96770) is the principal ortholog studied; LEF1/TCF biology is conserved broadly across vertebrates. (Finding F008.)
- **Natural disease / veterinary relevance:** no naturally occurring companion-animal or wildlife equivalent of ECTD17 is documented in OMIA for this gene/phenotype combination; relevance is through engineered models, not spontaneous animal disease.
- **Comparative biology:** The canonical WNT/β-catenin → TCF/LEF transcriptional axis is deeply evolutionarily conserved; LEF1-dependent ectodermal appendage development (hair, teeth, mammary) is conserved between mouse and human. However, species differences in WNT/TCF-LEF usage exist — e.g., *"clear differences between mouse and human pluripotent stem cells"* in TCF/LEF gene regulation (PMID: 40844425) — cautioning against assuming identical requirements across contexts.
- **Transmission / zoonosis:** not applicable.

Section 15 — Model Organisms

Primary model: *Lef1*-null mouse (MGI:96770). Alliance of Genome Resources / MGI curated phenotypes (51 annotations) include: absent teeth/incisors/molars and arrested tooth development; abnormal/underdeveloped hair follicles, decreased hair follicle number, alopecia; absent/sparse vibrissae; absent mammary gland and abnormal mammary bud;

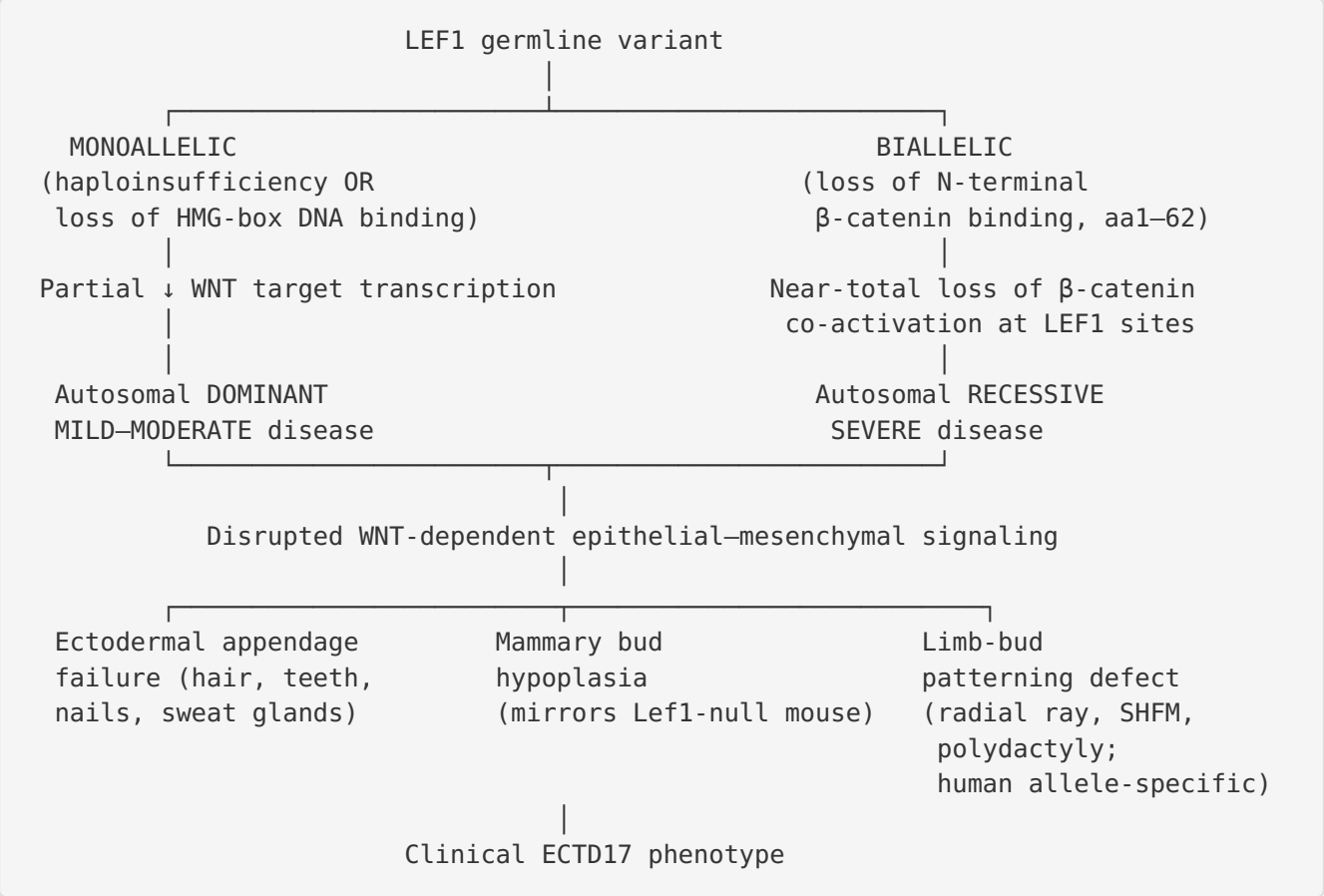
abnormal skin; abnormal craniofacial morphology; dilated cardiomyopathy; absent dentate gyrus/hippocampus and absent cortical barrels; increased skin-tumor incidence; and perinatal/postnatal/preweaning lethality with complete penetrance. (Finding F008.)

Phenotype recapitulation and the key allele/dosage insight. The null mouse **recapitulates the ectodermal and mammary components** of the human disease (teeth, hair, whiskers, mammary gland), confirming LEF1-dependence of these appendages: LEF-1 is expressed in *"the dermal papilla of hair and whisker follicles, the dental papilla of teeth, and a subset of mesenchymal cells in pubescent mammary glands"* (PMID: 9786416). Crucially, the null mouse does **not** list limb reduction / split-hand-foot / polydactyly phenotypes, and complete knockout is perinatally/postnatally lethal — whereas human patients survive and manifest limb malformations. This dissociation indicates that: 1. the **limb phenotype in humans is allele- and dosage-specific** (partial/domain-selective losses or heterozygous haploinsufficiency), not reproduced by total gene ablation; and 2. mouse complete-null lethality and additional phenotypes (CNS, cardiomyopathy) are **model limitations** not seen in human patients. (Finding F008.)

Genetic model types and resources. Conventional *Lef1* knockouts are the main resource (MGI/IMSR). Conditional/tissue-specific and hypomorphic/knock-in alleles that better mimic human domain-selective variants (e.g., β-catenin-binding–deficient or HMG-box–truncated alleles) would more faithfully model the genotype–severity relationship. In vitro systems (dermal papilla cells, hair-follicle organ culture, reporter assays) are well established WNT/LEF1 readout models (e.g., PMID: 42283394, PMID: 42316873).

Mechanistic Model / Interpretation

The investigation converges on a single, internally consistent model in which **a WNT-effector transcription factor's dosage and domain integrity map directly onto inheritance mode and clinical severity**:



Three independent lines of evidence reinforce this model: (1) the **functional dissection** by Dufour et al. tying variant class to mechanism and severity; (2) the **population-genetics constraint** (pLI 0.998) confirming LEF1 as haploinsufficiency-sensitive; and (3) the **protein domain map** (UniProt) that physically separates the two loss-of-function mechanisms into the N-terminal β-catenin-binding domain and the C-terminal HMG box. The **mouse model** both validates the ectodermal/mammary arm of the mechanism and, by its failure to reproduce the limb phenotype under complete null conditions, sharpens the interpretation that the human limb malformations arise from specific partial/heterozygous losses rather than total absence of LEF1.

Evidence Base

PMID	Title (abbrev.)	Role in this report
35583550	*Monoallelic and biallelic variants in LEF1 … new syndrome combining ectodermal dysplasia and limb malformations caused by altered WNT signaling*	**Primary/defining paper.** Establishes causality, the two-mechanism genotype–severity model, phenotype spectrum, and altered WNT signaling (RNA-seq).
9786416	LEF-1 expression at epithelial–mesenchymal interaction sites	Confirms LEF1 dependence of hair/whisker, tooth, mammary appendages — the ectodermal arm of the mechanism.
34573371	Ectodysplasin pathway key players in ED	Establishes WNT (and EDA) pathway disruption as an established ED cause, contextualizing LEF1 as a WNT effector.
42283394	ROBO2 marker / DPC culture with WNT+FGF	LEF1 is a WNT-responsive readout (3.8-fold induction, $p<0.01$) in dermal papilla cells.
42316873	Sphingomonas EVs and hair growth	LEF1/AXIN2 induced as canonical Wnt readouts in hair follicles.
30622331	10q24 duplication / SHFM spectrum	Epidemiologic anchor for the SHFM component (1/8,500–25,000).
31420900	TP63 variant, SHFM + hypodontia	Key differential: TP63-related SHFM + hypodontia.
31332306	SHFM molecular studies (LRP6, UBA2, WNT10B, TP63, DLX5)	Differential-diagnosis gene set for the limb phenotype.
42110406, 33294441	Novel TP63 variants in SHFM	Additional TP63 differential evidence.
40844425	TCF/LEF in human vs rodent pluripotency	Cautions on mouse–human differences in TCF/LEF requirement.

Peripheral LEF1 literature (cancer, T-cell memory, hair-growth pharmacology, β-catenin condensates) was reviewed to characterize LEF1 biology but does **not** bear directly on ECTD17 pathogenesis; it is retained only as supporting context for LEF1's role as a WNT/β-catenin transcriptional effector.

Limitations and Knowledge Gaps

1. **Single-cohort evidence.** Essentially all disease-specific clinical/molecular evidence derives from one report of 12 individuals (PMID: 35583550). Phenotype frequencies (e.g., split hand 3/12) are small-n and uncertain.
2. **No epidemiology.** Prevalence, incidence, penetrance, sex ratio, and demographic distribution are unknown.
3. **Genotype–phenotype resolution.** The two-mechanism model is well supported but based on limited variants; the full allelic series (especially missense variants affecting intermediate functions) is not yet mapped to outcomes.
4. **Limb-mechanism gap.** The precise WNT/LEF1 circuitry producing radial-ray/SHFM defects in humans is not experimentally resolved, and the mouse null fails to model it — leaving the limb causal step partly inferred.
5. **No molecular profiling beyond the founding study.** No independent transcriptomic, proteomic, epigenomic, or single-cell datasets specific to patient tissues exist.
6. **No therapeutics.** No disease-specific pharmacology, trials, or outcome data; QoL has not been formally measured.
7. **Model limitations.** Existing *Lef1*-null mice carry confounding phenotypes (lethality, CNS, cardiomyopathy) and do not recapitulate the limb phenotype; domain-selective/conditional alleles are needed.

Proposed Follow-up Experiments / Actions

1. **Expand the patient registry.** Aggregate additional cases via GeneMatcher/Matchmaker Exchange to refine phenotype frequencies, penetrance, expressivity, and the allelic series, and to establish epidemiology.

2. **Functional allelic series.** Systematically test patient and engineered *LEF1* variants in parallel β-catenin-binding (pulldown), DNA-binding (HMG-box), and WNT-reporter/RNA-seq assays to place each variant on the dosage–severity axis (extending the Dufour framework).
3. **Domain-selective mouse models.** Generate knock-in alleles that selectively ablate β-catenin binding (aa 1–62) vs truncate the HMG box, plus limb-bud–conditional alleles, to test whether human-like domain-selective losses reproduce the limb phenotype that the complete null does not.
4. **Developmental limb circuitry.** Use limb-bud organoids / single-cell and spatial transcriptomics to define LEF1-dependent WNT targets in radial-ray and AER/ZPA programs, directly testing the inferred limb causal step.
5. **Variant curation.** Reclassify the "conflicting" p.Leu157fs and VUS p.Glu271Ter with functional data; apply PVS1/PM2 reasoning given extreme LoF intolerance.
6. **Clinical care pathway.** Develop a multidisciplinary surveillance protocol (dental, orthopedic/hand, dermatologic, cardiac, neurodevelopmental) and apply standardized QoL instruments to capture morbidity.
7. **Genetic counseling tooling.** Codify variant-class-based recurrence-risk counseling (AD vs AR) and consanguinity risk for the severe recessive form.

Report compiled from 8 confirmed findings across 5 investigation iterations and 28 reviewed papers. Evidence source types: human clinical/molecular (Dufour et al. cohort; ClinVar; gnomAD), model organism (MGI Lef1*-null), in vitro (WNT/LEF1 readout systems), and computational/ontology curation (MONDO, OMIM, UniProt, HPO, GO).*
